

Palette-based Image Search with Color Weights

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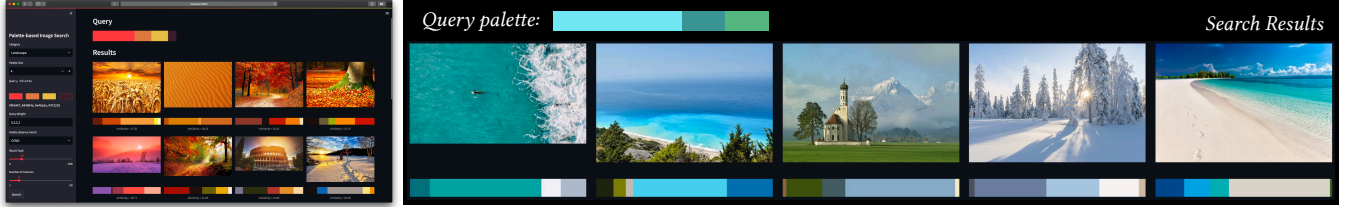


Figure 1: We propose a palette-based image search system that can help users' visual design exploration with query palettes.

ABSTRACT

We propose a novel image search system with color palettes. By querying color palettes, users can search for inspiring images, which is helpful for design exploration. Although few such systems can accept palettes as queries, they have several constraints on query palettes (e.g., limited palette size) or search results. Our system accepts palettes with color weights and any sizes as the inputs and returns results with enough diversity that can stimulate users' design inspiration. To achieve these, we extract color themes from our database with a perceptually-based model and calculate the palette similarity based on a weighted palette distance. Visual comparisons demonstrate that our system has more potential for helping users' design inspiration than the existing systems.

CCS CONCEPTS

• Information systems → Image search.

KEYWORDS

Color palette, color theme, palette similarity, image search

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1 INTRODUCTION

Color is one of the essential components of visual design. Most color design involves multiple colors, i.e., color combinations to achieve

visually compelling compositions. However, designing color combinations is a demanding task even for professional designers since there are *good* or *not good* color combinations, i.e., color harmony. This is also true for novices because most people face color design tasks, such as preparing presentation slides, fashion, or room decoration. Although we can refer websites such as Adobe Color [Adobe 2022], there are limited samples and/or functionality to explore inspiring visual designs. Especially, it is difficult for searching with color combinations (palettes). Art Palette [Google 2022] provides functionality to search arts by query palettes. However, the palette size is limited to five colors, which restricts flexible search. Multicolor [Labs 2022] can search Flickr images by querying color palettes (up to five colors). It also accepts color weights and tags. While these functionalities are helpful when searching for desired images, the results can be redundant, i.e., many similar images can be listed owing to searching on Flickr, which is not suited for visual design exploration. In this paper, we present a novel palette-based image search that can accept query palettes comprising any colors with weights (Fig. 1). By building our database collected from Unsplash [Unsplash 2022], we can search for visually inspiring images. We have two main contributions: (1) we extract color themes from our dataset based on [Lin and Hanrahan 2013] that outperforms the widely used K-means variants to better capture how people extract color themes from images; and (2) we propose an inter-palette distance measure with weights and without palette size restrictions. We demonstrate our method and find that our system has the potential to help users' visual design exploration.

2 OUR APPROACH

System overview. A user can search images in the database by querying a color palette $\mathbf{P} = [\mathbf{c}_1 \ \mathbf{c}_2 \ \dots \ \mathbf{c}_n]^\top$ with color weights $\mathbf{w} = [w_1 \ w_2 \ \dots \ w_n]^\top$, where $\mathbf{c}_i$ is a color with its weight w_i . A category T and the number of results K are also specified. The query is sent to the server and the feature matrix is calculated; $\mathbf{F}_{\text{query}} = [\mathbf{P} \mid \mathbf{w}]$. Then, we calculate the feature distance, i.e., palette similarity, and the top K images with those palettes similar to the query are returned to the frontend and displayed to the user.

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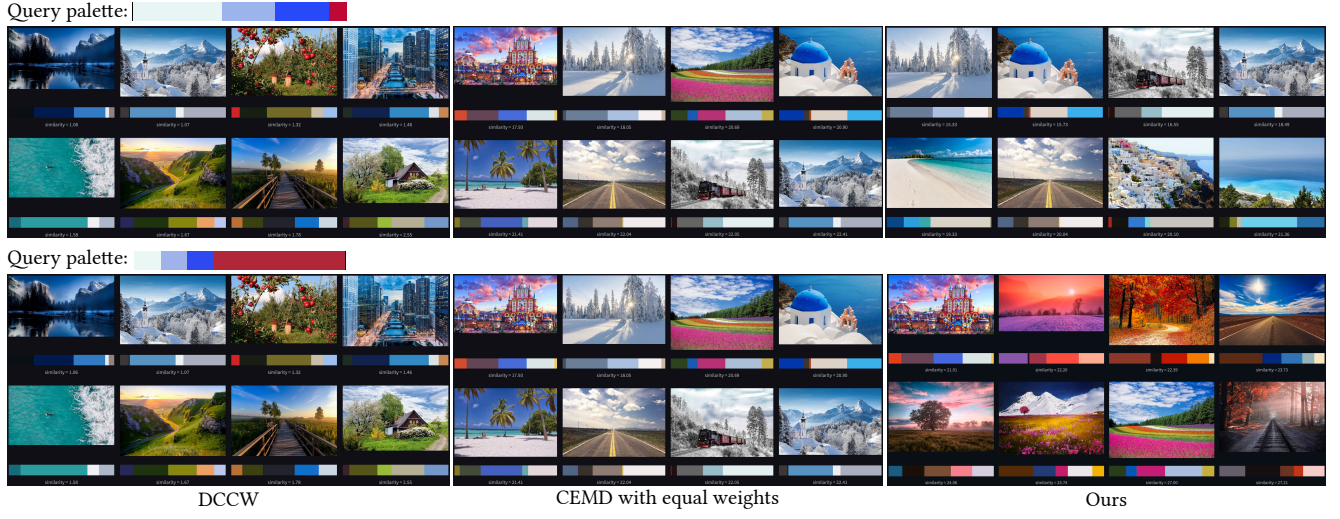


Figure 2: Search results of two palettes with different weights in the “Landscape” category. Left to right: DCCW [Kim and Choi 2021], CEMD with equal weights [Skaff et al. 2011], and ours.

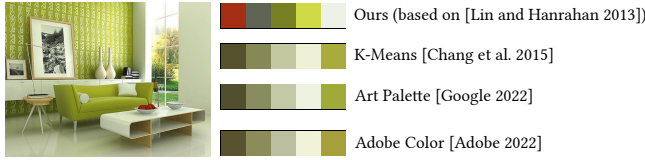


Figure 3: Comparison of color theme extraction methods. Left: target image. Right: extracted palettes.

Building database. We build our database by collecting inspiring images I_i from Unsplash [Unsplash 2022]. As a preprocessing, we extract color palettes from the collected images based on Lin and Hanrahan’s model (LH) [Lin and Hanrahan 2013]. For convenience, we upgrade the saliency estimation to a transformer model [Lou et al. 2022]. Although we could not find any significant accuracy improvement, it is more convenient to work with than the original model with many external dependencies. Fig. 3 shows a comparison of color theme extraction methods. Only our LH-based model captures the *accent* color derived from red apples on a table. We store the images I_i , extracted palettes P_i , and color weights w_i in our database. We compute the similarity, i.e., palette distance to find the images that are recalled by the corresponding palettes.

Palette distance measure. We compare F_{query} and feature matrices $F_i \in T$ in the database. To measure the similarity (distance), we employ a variant of the Earth Mover’s Distance with CIE Delta E 2000 as the color distance measure similar to the Color-based EMD (CEMD) [Skaff et al. 2011]. Different from existing works [Kim and Choi 2021; Skaff et al. 2011], we compute color weights used in the CEMD by finding palette color assignments for each image pixel and counting the number of palette colors assigned. To recap, our problem is formulated as the following image search problem:

$$\text{find } I_{i^*} \text{ such that } i^* = \arg \min_{i \in |T|} \text{CEMD}(F_{\text{query}}, F_i).$$

3 RESULTS

We sampled 60 “Landscape” images from our database for evaluation. We compared our search results with DCCW [Kim and Choi 2021] and CEMD with equal weights [Skaff et al. 2011]. Figure 2 shows the search results of two palettes with different weights. Our weighted distance measure better captures the *dominant* and *accent* colors than that of other methods. The bottom row shows the failure cases of the unweighted approaches. Without a weighted distance measure, the dominant color (red) is ignored and the results include a lot of *cool* images. In contrast, our results successfully capture the dominant color and find *warm* images. DCCW is one of the state-of-the-art methods; however, our experiments show that the results seem not convincing in our purpose; i.e., inspiring visual design exploration. Moreover, the computation is not fast enough for interactive exploration; our CEMD-based approach is ~ 10 times faster. In the future, we plan to conduct more dedicated experiments to improve our system more practical.

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